SUPPORTING ADULT GUIDANCE

Sine and cosine

Introduction to Trigonometry. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner meets the **sine** and **cosine** ratios and organises all three trigonometric ratios as **SOH · CAH · TOA**: sin θ = opposite ÷ hypotenuse, cos θ = adjacent ÷ hypotenuse, tan θ = opposite ÷ adjacent. The emphasis is on **selecting the correct ratio from the sides involved**, on the behaviour of sine and cosine (one rises as the other falls; both lie between 0 and 1), and on the **complementary relationship** sin θ = cos(90° – θ).

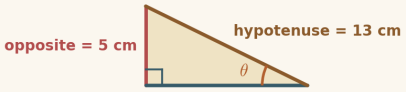
Driving Question: *How do the other two ratios complete the picture? (SOH · CAH · TOA)*

The mathematics, in plain terms

This lesson completes the trio of trigonometric ratios. The **sine** of an angle is **opposite ÷ hypotenuse** (SOH) and the **cosine** is **adjacent ÷ hypotenuse** (CAH); together with the tangent from Lesson 3 (opposite ÷ adjacent, TOA) they are held in the single map **SOH · CAH · TOA**. The key new feature is that **sine and cosine bring in the hypotenuse**, which the tangent never uses. Because a side of a right-angled triangle can never be longer than the hypotenuse, both sine and cosine always lie **between 0 and 1**. As the angle increases, the opposite side lengthens and the adjacent side shortens, so **sin θ increases while cos θ decreases** — they move in opposite directions. The most elegant idea in the lesson is the **complementary relationship**: the side opposite one acute angle is adjacent to the other (the swap first seen in Lesson 2), so the **sine of one acute angle equals the cosine of the other**, and since the two acute angles add to 90°, **sin θ = cos(90° – θ)**. The single skill that matters most now is **choosing the right ratio** from the sides in play — which is exactly what the side-finding and angle-finding lessons that follow will depend on. The common error to watch for is **confusing sine and cosine** (which of opposite or adjacent goes on top); reading the mnemonic carefully — S-O-H, C-A-H — heads it off.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

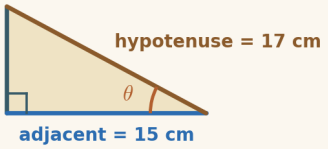


Example 1 — finding the sine

The sine of the marked angle is the opposite divided by the hypotenuse. Here the opposite is 5 cm and the hypotenuse is 13 cm.

1. sine uses opposite and hypotenuse	$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$
2. put in the numbers	$\sin\theta = \frac{5}{13}$
3. work it out	$\sin\theta \approx 0.385$

sin θ ≈ 0.385



Example 2 — finding the cosine

The cosine of the marked angle is the adjacent divided by the hypotenuse. Here the adjacent is 15 cm and the hypotenuse is 17 cm.

1. cosine uses adjacent and hypotenuse	$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
2. put in the numbers	$\cos\theta = \frac{15}{17}$
3. work it out	$\cos\theta \approx 0.882$

cos θ ≈ 0.882

Three ratios, one structure

The heart of this lesson is turning three separate facts into **one structure**. SOH · CAH · TOA is not three things to memorise but a single map in which each name tells you exactly which two sides to use. It is worth being explicit that sine and cosine **both use the hypotenuse** while the tangent does not — that is the feature that distinguishes the new ratios from the familiar one. Two further ideas deepen it. First, **behaviour**: grounding sine and cosine in the moving triangle (opposite up, adjacent down, so sine up and cosine down; both between 0 and 1 because no side beats the hypotenuse) makes them feel inevitable rather than arbitrary. Second, **symmetry**: letting the learner discover that $\sin\theta = \cos(90° - \theta)$ — by computing the sine of one angle and the cosine of the other on the same triangle — reveals that sine and cosine are really one ratio read from the two different angles. Securing the habit of **naming the ratio from the sides involved** is the payoff that the next lessons build on directly.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Ratio Trio: on a 3–4–5 triangle, light up the two sides each ratio uses and read its value — SOH · CAH · TOA.	~14 min
Movement	Ratio Explorer: with the hypotenuse fixed, watch sin θ rise and cos θ fall as the angle grows.	~12 min
Reflection	Which ratio is it?: name the ratio (sine, cosine or tangent) from the two highlighted sides.	~10 min
Creativity	The other angle's cosine: find that the sine of one acute angle equals the cosine of the other.	~6 min
Rest	A pause after seeing three ratios fold into one tidy structure.	~3 min
Nutrition	Mode B (intellectual): what the learning feeds — organising facts into a memorable structure, and noticing hidden symmetry.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Confusing sine and cosine (which side goes on top)

Repair: “Read the mnemonic: S-O-H means Sine = Opposite over Hypotenuse; C-A-H means Cosine = Adjacent over Hypotenuse.”

Using the hypotenuse in the tangent

Repair: “Tangent is TOA — opposite over adjacent. Only sine and cosine use the hypotenuse.”

Thinking sine or cosine can be more than 1

Repair: “Can a side be longer than the hypotenuse? No — so opposite ÷ hypotenuse and adjacent ÷ hypotenuse are always at most 1.”

Forgetting which two sides a ratio needs

Repair: “Name the two sides first, then choose: opposite & hypotenuse → sine; adjacent & hypotenuse → cosine; opposite & adjacent → tangent.”

Not seeing that sine and cosine are linked

Repair: “Work out the sine of one acute angle and the cosine of the other on the same triangle. What do you notice?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: naming the opposite, adjacent and hypotenuse relative to a marked angle (Lesson 2), the tangent ratio $\tan\theta$ = opposite ÷ adjacent (Lesson 3), and a ratio written as a division. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the structure and the two big ideas (behaviour, symmetry) with the learner:

- “Which two sides does sin θ use? And cos θ? And tan θ?”
- “As the angle got bigger, what happened to sin θ? And to cos θ?”
- “Why can sine and cosine never be bigger than 1?”
- “Work out sin of one acute angle and cos of the other — are they equal? Why?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“What is the difference between sine and cosine?”

Sine is opposite ÷ hypotenuse; cosine is adjacent ÷ hypotenuse. They use the same hypotenuse but a different second side, and they are linked: the sine of one acute angle equals the cosine of the other.

“Why do sine and cosine never go above 1?”

Because a side of a right-angled triangle can never be longer than the hypotenuse. Dividing a shorter side by the hypotenuse always gives a number between 0 and 1.

“Do I have to memorise three separate formulas?”

No — SOH · CAH · TOA is one structure. Each group of three letters tells you the ratio and the two sides it uses, so you can rebuild any of them from the mnemonic.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.